

REDWOOD FALLS, MN, USA

WMO: 726556

Lat: 44.548N

Lon: 95.080W

Elev: 311

StdP: 97.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-22.6	-28.8	0.3	-24.7	-26.3	0.4	-22.2	12.5	-9.9	11.4	-8.8	4.3	290	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.9	23.4	31.3	22.6	29.8	21.5	25.4	30.8	24.2	29.0	23.1	27.8	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.4	28.6	22.6	18.0	27.4	21.5	16.8	26.3	79.6	30.7	74.7	29.1	70.2	27.7	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	8.9	DB	-28.6	36.2	3.3	1.4	-31.0	37.2	-32.9	38.0	-34.8	38.7	-37.2	39.7	(4)
(5)			WB	-28.7	27.3	3.3	1.3	-31.1	28.2	-33.0	29.0	-34.9	29.7	-37.3	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.7	-7.4	-0.3	7.8	14.9	20.6	22.9	21.3	17.2	9.4	1.0	-6.6	(6)	
	DBStd	12.75	7.49	7.24	7.17	5.70	4.91	3.69	3.13	3.03	4.65	5.35	6.30	6.75	(7)	
	HDD10.0	2412	610	487	324	106	13	0	0	0	3	77	276	516	(8)	
	HDD18.3	4293	868	721	577	318	130	18	3	8	75	280	521	774	(9)	
	CDD10.0	1562	0	0	6	39	164	319	398	349	221	59	6	0	(10)	
	CDD18.3	402	0	0	0	2	23	88	144	99	42	4	0	0	(11)	
	CDH23.3	3873	0	0	2	37	293	840	1386	829	424	59	2	0	(12)	
	CDH26.7	1323	0	0	0	9	102	294	512	254	136	16	0	0	(13)	
(14) Wind	WSAvg	4.5	4.8	4.9	4.8	4.9	4.8	4.2	3.6	3.4	4.1	4.7	4.8	4.6	(14)	
(15) Precipitation	PrecAvg	679	17	18	39	67	81	102	98	81	62	51	36	17	(15)	
	PrecMax	954	49	43	110	192	244	293	188	211	172	179	135	58	(16)	
	PrecMin	382	0	0	7	8	12	22	13	22	8	1	0	1	(17)	
	PrecStd	160	12	11	24	39	46	55	49	52	41	41	32	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	12.5	21.6	27.5	33.1	34.9	35.2	33.4	33.0	29.0	20.6	10.9	(19)	
		MCWB	3.9	8.5	13.4	16.0	19.6	22.9	25.9	25.2	21.7	17.4	12.6	8.1	(20)	
	2%	DB	3.9	7.4	16.6	23.4	29.4	32.1	32.9	31.3	30.2	24.5	16.5	6.8	(21)	
		MCWB	1.6	4.9	11.4	13.5	17.9	22.2	24.5	23.7	21.1	15.9	11.0	3.8	(22)	
	5%	DB	1.9	4.2	12.8	20.4	26.6	30.0	31.2	29.5	27.9	21.1	12.8	3.7	(23)	
		MCWB	0.5	2.3	9.2	11.8	16.9	20.7	23.7	22.6	19.8	14.2	8.2	1.9	(24)	
	10%	DB	0.4	2.3	9.3	17.4	23.8	27.9	29.5	27.7	25.8	18.0	9.8	1.9	(25)	
		MCWB	-0.6	1.0	6.6	10.3	15.3	19.7	22.3	21.3	18.6	12.7	6.3	0.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	8.8	15.5	17.2	22.2	25.1	27.6	26.8	24.2	20.0	14.6	7.8	(27)	
		MCDB	7.8	12.0	18.6	24.4	29.0	31.7	32.3	31.7	29.8	24.7	19.1	9.8	(28)	
	2%	WB	1.9	4.9	12.1	15.0	19.9	23.7	25.8	24.9	22.7	17.3	11.2	3.8	(29)	
		MCDB	3.5	7.4	15.7	21.2	26.4	29.6	31.2	29.6	27.7	22.4	15.4	6.2	(30)	
	5%	WB	0.7	2.6	9.5	13.0	18.2	22.3	24.7	23.6	21.4	15.2	8.8	2.0	(31)	
		MCDB	1.8	4.2	12.8	18.4	24.3	28.0	29.7	27.9	26.0	20.2	12.5	3.5	(32)	
	10%	WB	-0.5	0.9	6.5	11.3	16.6	21.1	23.6	22.4	19.9	13.2	6.4	0.6	(33)	
		MCDB	0.5	2.2	9.5	16.5	22.1	26.4	28.1	26.4	24.0	17.6	9.8	1.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	9.2	11.1	11.7	11.1	11.4	11.3	12.5	11.2	9.5	8.3	(35)	
		MCDBR	10.2	11.2	14.4	16.9	16.3	13.8	12.9	13.0	15.2	16.2	14.8	11.1	(36)	
	5% WB	MCWBR	8.4	8.7	9.5	9.2	7.7	6.2	6.3	6.4	7.2	8.7	9.6	8.8	(37)	
		MCWBR	9.6	10.6	13.8	14.1	13.7	11.9	11.7	11.6	12.5	13.9	13.7	10.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.260	0.277	0.306	0.362	0.384	0.399	0.406	0.410	0.367	0.320	0.281	0.262	(40)	
		taud	2.470	2.425	2.457	2.341	2.342	2.357	2.320	2.286	2.383	2.511	2.541	2.474	(41)	
	Edn at Noon	Edn at Noon	881	922	935	899	886	871	859	841	853	859	849	841	(42)	
		Edh at Noon	68	86	96	118	122	121	124	124	104	79	64	62	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.84	2.87	3.79	4.69	5.42	6.04	6.53	5.39	4.23	2.77	1.80	1.36	(44)	
	RadStd	RadStd	0.20	0.34	0.38	0.47	0.41	0.45	0.22	0.43	0.31	0.36	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page